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To cite this article: Weijin Qin *et al* 2024 *Phys. Scr.* **99** 125026

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Analysis of the performance of the PPP-B2b satellite clock and its effects on time transfer

RECEIVED
3 August 2024REVISED
5 November 2024ACCEPTED FOR PUBLICATION
8 November 2024PUBLISHED
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Keywords: BDS-3, PPP-B2b, satellite clock performance, time transfer

Abstract

BDS-3 is capable of providing precise point positioning (PPP) services via B2b signals for the Asia-Pacific region. The performance of the satellite clock plays a dominant role in the transmission interval of B2b clock correction and PPP time transfer. In this work, first, the coverage of B2b signals is analysed, and the performance of the satellite clock is evaluated (including orbit accuracy, clock stability, clock accuracy and prediction accuracy). Second, the results of time transfer based on BDS-3 PPP-B2b are investigated. The values show that the orbit error of BDS-3 recovered from PPP-B2b is better than that of the GPS recovered from PPP-B2b. The stability of BDS-3 recovered from the PPP-B2b clock is superior to that of the GPS PPP-B2b clock. The stability of the BDS-3 clock is approximately $1\sim 2 \times 10^{-13}$, $5\sim 9 \times 10^{-14}$, and $3\sim 5 \times 10^{-14}$ at 100 s, 1000 s, and 10000 s, respectively. No obvious discrepancy exists in the BDS-3 clocks of different types. The mean prediction accuracy of BDS-3 is 0.03 ns when the prediction time is up to 120 s, and the prediction accuracy of the GPS is 0.28 ns when the prediction time is merely 30 s. This result shows that it is possible to decrease the transmission interval of BDS-3 B2b clock correction and save much space. The time transfer based on BDS-3 PPP-B2b can subsequently achieve a 0.50 ns level. The frequency stability of the time link is approximately 9.25×10^{-14} , 2.42×10^{-14} , and 9.34×10^{-15} at 100 s, 1000 s, and 10000 s, respectively. We can conclude that the BDS-3 PPP-B2b solution is reliable for time transfer.

1. Introduction

The precise point positioning (PPP) technique has demonstrated long-term advantages in terms of time transfer because of its wide-area coverage, low cost and high accuracy [1, 2]. By virtue of the precise products calculated by the international GNSS service (IGS), the accuracy of time transfer at the subnanosecond level can be achieved. With the demand of PPP users with low latency, IGS real-time service (RTS) was established in 2001 [3]. Real-time orbit and clock state space corrections (SSR), which refer to broadcast navigation data, have been provided for recovering precise products. RTS uses Networked Transport of RTCM via Internet Protocol (NTRIP) for streaming GNSS observations and SSR over the internet [4]. SSRs are available with a sampling rate of 5 s of GPS/Galileo/BDS/GLONASS and are expressed within the International Terrestrial Reference Frame 2020 (ITRF20) [5]. Two sets of products are provided: the antenna phase centre and the centre of mass. Therefore, the PPP technique provides processing capacity in real-time mode.

As we expected, many researchers have focused on real-time time transfer. Zhang *et al* proved that GPS/Galileo/BDS/GLONASS single-system time transfer can reach the subnanosecond accuracy level [6]. Guo *et al* reported that the stability of the 1PPS output of the GNSS one-way timing system is better than that of the 1 ns output of one day [7]. Ge *et al* emphasized that RT-PPP time transfer with different real-time products can achieve 0.3 ns compared with the IGS final clock products [8]. Lyu *et al* reported that the variation in frequency is greater than 6.1×10^{-16} and that clock jumps are greater than 0.6 ns, both of which can be observed in GRE (GPS AR) RT PPP

Table 1. PPP-B2b satellite information.

PRN	SVN	Satellite type	Launch date	Satellite status	Service signal	Position
59	GEO-01	BDS-3	2018.11.01	operational	B2b	140°E
60	GEO-02	BDS-3	2020.03.09	operational	B2b	80°E
61	GEO-03	BDS-3	2018.11.01	operational	B2b	110.5°E
62	GEO-04	BDS-3	2020.03.09	testing	B2b	160°E

mode [9]. Ge verified that BDS-3 real-time PPP time transfer can reach 0.02 ns with respect to the smoothed residuals [10].

SSR, as the essentiality of RT PPP processing, has been transmitted through the internet; unfortunately, its application has been restricted by internet facilities, such as network outages or desert/marine/mountain areas [11, 12]. Satellite-based PPP augmentation services have gradually been constructed to address the deficiency of internet-based services. The centimetre-level augmentation service (CLAS) of the QZSS, the PPP-B2b service of the BDS and the high-accuracy service (HAS) of Galileo have finished [13, 14]. CLAS can provide centimetre-level positioning services for QZSS, GPS and Galileo users in the Japanese area via the L6 signal [14]; PPP-B2b provides centimetre-level kinematic positioning for GPS and BDS-3 users in the Asia-Pacific area via the B1I signal [15]; and Galileo High Accuracy Service (HAS) can transmit corrections for GPS and Galileo users in the global area via the E6b signal. Hence, PPP-B2b is a unique way to obtain BDS-3 corrections from satellite-based augmentation services.

The performance of PPP-B2b time transfer has been analysed by several researchers. Zhang conducted a BDS-3 B1I/B3I PPP-B2b time transfer experiment with a zero baseline and long baseline. Ge compared BDS-3 B1I/B3I PPP-B2b time transfer results with different kinds of products [16]. Tang verified that the accuracy of BDS-3 PPP-B2b time transfer with DCB correction is improved by approximately 64% compared with that without DCB correction [17]. Zhang claimed that the uncertainty of UTC dissemination with BDS-3 PPP-B2b is better than 0.7 ns [18].

With the enhancement of the technique and manufacture of on-board clocks, the performance of on-board clocks has been greatly improved [19, 20]. This provides the opportunity to decrease the transmission interval of the PPP-B2b satellite clock on the basis of ensuring service quality. The paper starts with an analysis of the coverage of the PPP-B2b service. Then, the short-term frequency stability and forecasting performance of the PPP-B2b satellite clock are evaluated. A solution for PPP-B2b time transfer was subsequently introduced, and an experiment was conducted. The conclusions are presented in section 4.

2. Analysis of the coverage of the PPP-B2b service

A prerequisite of PPP-B2b service is that GEO satellites broadcast by PPP-B2b corrections can be observed at the user location. Considering more guidelines for PPP-B2b users, it is necessary to analyse the relationship between the location of the user and the distribution of GEO satellites. As shown in table 1, the service signal of B2b has been transmitted routinely by three BDS-3 GEO satellites. In this study, the GBM final products provided by the GFZ analysis centre are used to analyse the visibility of the GEO satellite on DOY 351 in 2023. In the current status, PRNs 59, 60 and 61 are operational, and PRN 62 is tested. To our knowledge, the ephemeris of C59 and C60 has been provided only in the GBM orbit file. The same correction values are transmitted with three different GEO satellites.

The Earth has been assumed to be a plane. It has been divided into multiple grids equally. The location of a user can be represented by each separated grid. The plane ranges from 180°W to 180°E and from 90°N to 90°S. The grids are separated into 5° latitude and longitude, and the Earth is divided into 2701 grids. Each grid has been divided into m and n. The expressions are as follows:

$$\text{lat} = 180/m \quad (1)$$

$$\text{lon} = 360/n \quad (2)$$

Figure 1 shows that B2b covers most of Asia, Oceania, Europe, and Africa. The four subplots demonstrate the coverage status at 0 s, 28800 s, 57600 s and 86400 s. The green area indicates that at least one satellite can be observed the whole day, and the red area indicates that both C59 and C60 can be observed the whole day. The total unviewable area ranges from 130°W to 0°W and from 90°S~90°N. Owing to the motionless characteristic of GEO satellites, the coverage areas of red and green are symmetric, and the coverage of other days can yield the same conclusion as DOY 351 and is not mentioned here. In figure 2, the percentage between the visible area and the total area is shown. From 0 to 86400 s, with an interval of 300 s, the average percentage is 75.75%, and the

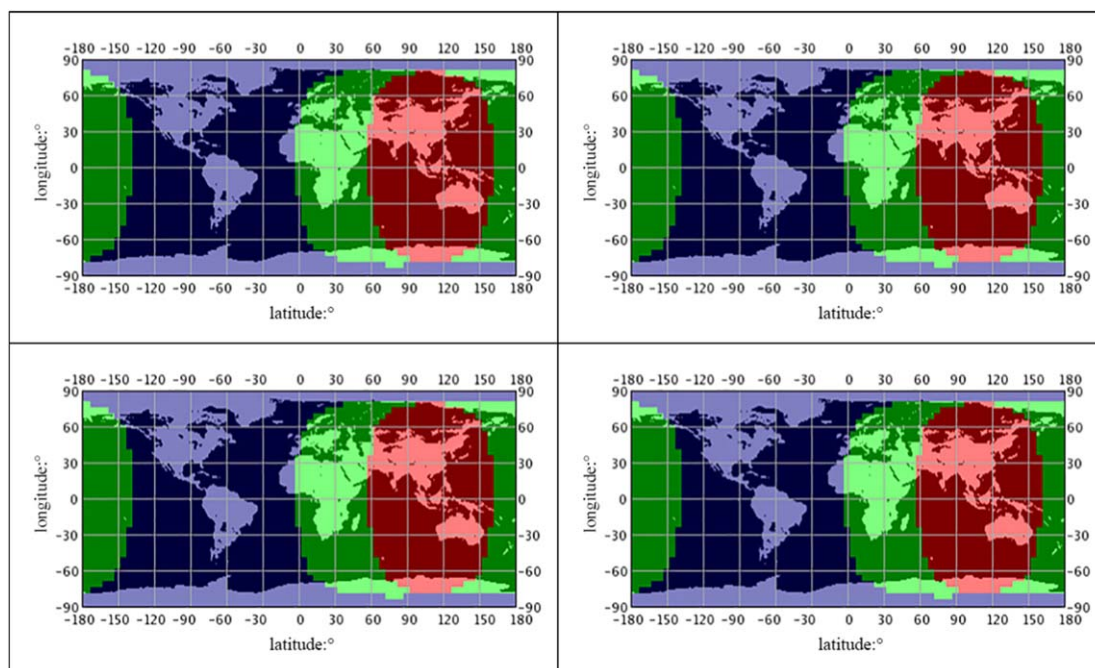


Figure 1. Ground coverage of C59 and C60 on DOYs 351 and 2023.

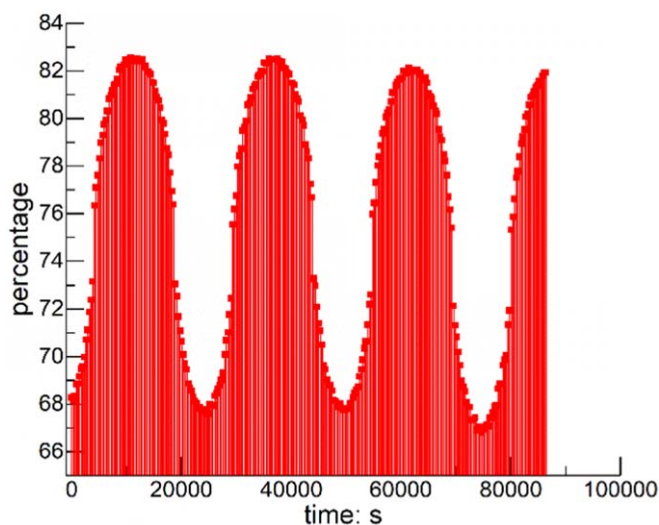


Figure 2. PPP-B2b visible percentage on DOY 351, 2023.

maximum percentage and the minimum percentage are 82.56% and 66.79%, respectively. We can draw two conclusions: (1) it is continuous in some areas; (2) as we expected, more than half of the users in China and surrounding areas can obtain PPP-B2b services; and (3) the percentage presents a periodic variation, and the cycle is approximately 24900 s, which is attributed to the effect on the period variation of the GEO orbit.

3. Recovery of the PPP-B2b orbit and clock offset

The recovery of the PPP-B2b orbit and clock offset is necessary prior to offering PPP service. The PPP-B2b precise orbit has been recovered with the combination of broadcast ephemeris and RAC (radial, along, and cross) correction [21]. The expression of the orbit is as follows:

$$\begin{cases} e_r = \frac{r}{|r|} \\ e_c = \frac{r \times \dot{r}}{|r \times \dot{r}|} \\ e_a = e_c \times e_r \\ \Delta C = [\delta C_r, \delta C_a, \delta C_c] \\ \text{orb}_{\text{pre}} = \text{orb}_{\text{brd}} + (e_r, e_a, e_c) \cdot \Delta C \end{cases} \quad (3)$$

where orb_{pre} and orb_{brd} denote the precise ephemeris and the broadcast ephemeris, respectively; e_r , e_a , and e_c indicate the transformation matrix in the RAC direction; ΔC is the PPP-B2b correction in the RAC direction, which refers to the APC (antenna phase centre) and $\dot{r}$ are the direction unit vector of the satellite velocity and the direction unit vector of the satellite position, respectively, in the ECEF (Earth-centred, Earth-fixed). The interval of orbit correction is 48 s.

The PPP-B2b precise clock is recovered with the combination of broadcast ephemeris and clock offset correction. The expression of the clock is as follows:

$$\text{clk}_{\text{pre}} = \text{clk}_{\text{brdc}} - \frac{C_0}{c} \quad (4)$$

where clk_{pre} and clk_{brdc} denote the precise satellite clock and the broadcast satellite clock, respectively, and C_0 is the clock offset correction transmitted by PPP-B2b in metres. Notably, the broadcast ephemeris of BDS-3 and GPS is derived from CNAV1 and LNAV.

Prior to data processing, the bias relative to the signal tracking mode should be removed from the observation. The correction formula is shown here:

$$l_{\text{corr}} = l_{\text{raw}} - \text{DCB}_{\text{sig}} \quad (5)$$

where l_{corr} and l_{raw} denote the corrected observations and raw observations, respectively, and DCB_{sig} is the bias transmitted by PPP-B2b.

4. Analysis of the evaluation of the PPP-B2b product

The quality of PPP-B2b services highly relies on the precision of B2b products. PPP-B2b has been saved to the format of RINEX for orbit and clock comparison. The GBM precise product was chosen as a reference. We analysed the precision between the B2b corrections and the precise products over seven days from DOY (day of year) 351 to 357 of the year 2023. First, the B2b clock and orbit SSR correction are decoded. The decoding format is different for different receiver manufacturers. The SSR and C-NAV broadcasts are combined, and the sp3 and clock files are saved in RINEX format.

2.1 Orbit evaluation
The polyfitting of short-arc orbits will contaminate the orbit accuracy. To avoid influencing the orbit precision, the solution here is to polyfit the GBM orbit and interpolate the orbit at a sampling interval of 1 s. Later, at common epochs, the orbit difference of the specific satellite between the PPP-B2b product and the GBM product was converted to the R, A and C directions, and the orbit error (RMS) was calculated. The formula is as follows:

$$\text{RMS}_{\text{orbit}} = \sqrt{\frac{\sum_{i=1}^n \Delta \text{orbit}_i^2}{N}} \quad (6)$$

where N is the number of common epochs and orbit_i denotes the orbit difference of RAC.

Owing to space limitations, the RMSs of each BDS-3 and GPS satellite on DOYs 351 and 355 are displayed in figure 3. The RMSs for a single day and the mean RMSs for GPS and BDS-3 are shown in table 2.

Combining figure 3 and table 2, we obtain several findings:

- (1) In figure 3, the precision of the GPS orbit is obviously worse than that of the BDS-3 orbit. For the same navigation system, the orbit errors of specific satellites are close to each other.
- (2) For the statistics of 7 days, the mean RMS values of the BDS-3 orbit errors in the RAC directions are approximately 0.09 m, 0.62 m, and 0.57 m. Similarly, the mean RMS values of the GPS orbit errors in the RAC directions are approximately 0.14 m, 0.74 m, and 0.57 m. Overall, BDS-3 and GPS show the same tendency: the orbit error in the R direction is relatively smaller than that in the AC directions, and the orbit error in the A direction is the largest.
- (3) For the R and A directions, the precision of BDS-3 is better than that of the GPS. The enhancement is that ISL (intersatellite link) observations have been used as additional supplements in the orbit determination of

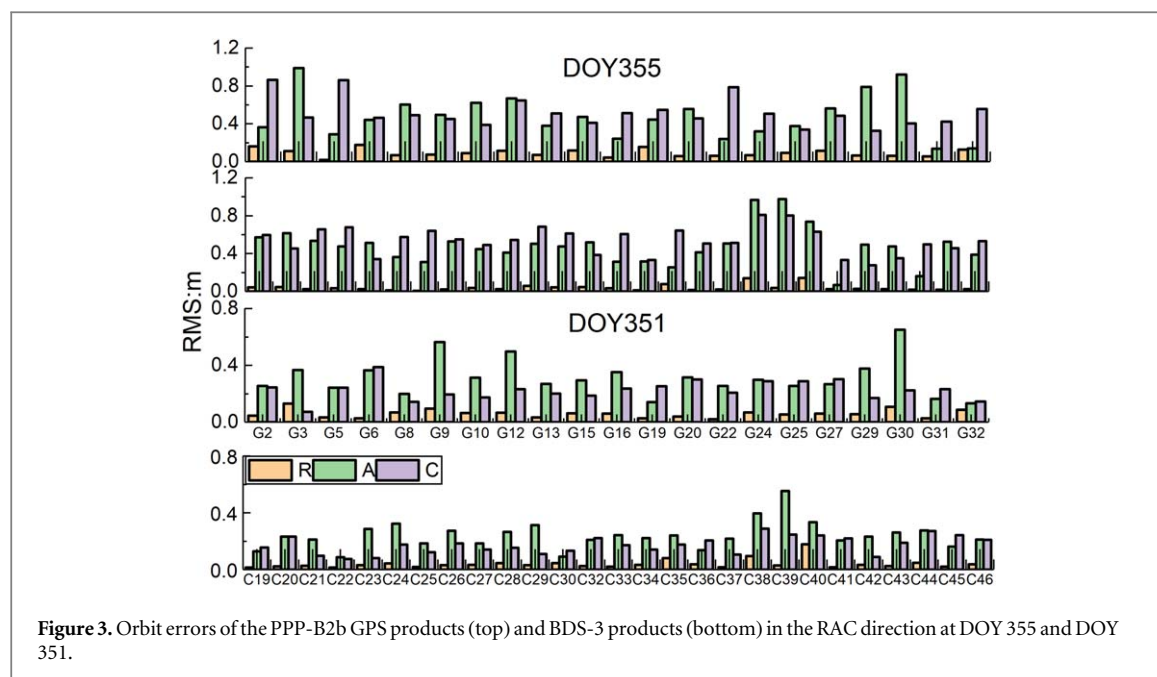


Figure 3. Orbit errors of the PPP-B2b GPS products (top) and BDS-3 products (bottom) in the RAC direction at DOY 355 and DOY 351.

Table 2. Mean RMS values of the PPP-B2b orbit errors.

RMS: m		DOY							MEAN
System		351	352	353	354	355	356	357	
BDS-3	R	0.08	0.08	0.06	0.08	0.08	0.20	0.07	0.09
	A	0.36	0.67	0.61	0.67	0.68	0.68	0.69	0.62
	C	0.24	0.55	0.61	0.55	0.62	0.70	0.79	0.57
GPS	R	0.14	0.14	0.14	0.14	0.16	0.15	0.14	0.14
	A	0.42	0.86	0.74	0.86	0.77	0.79	0.75	0.74
	C	0.29	0.61	0.56	0.61	0.62	0.62	0.69	0.57

BDS-3. In addition, the Asia-Pacific regional network rather than the global wide-area network has been applied for the estimation of PPP-B2b GPS products. The same conclusion was obtained from DOY 351 to 357. The RMS of the GPS/BDS-3 product is stable each day, which indicates the reliability and stability of the PPP-B2b orbit product.

4.1. Satellite clock evaluation

This section introduces the clock stability, clock prediction and clock precision. The Rb (rubidium) and PHM (passive hydrogen maser) clocks used by the BDS-3 satellites are homemade and are from the China Aerospace Science and Technology Corporation (CASC) and the Shanghai Engineering Center for Microsatellites (SECM). Considering that the short-term performance of satellite clocks is related to the type of atomic clock, BDS-3 and GPS satellites have been evaluated by different groups. More details for the BDS-3 and GPS satellites are listed in table 3.

4.1.1. Satellite clock stability assessment

The stability of the satellite clock plays an important role in the GNSS position/navigation/timing service. The Allan deviation has been selected for describing the stability because it converges for most types of clock noise. Prior to the stability analysis, the BDS-3/GPS clock offset recovered from the PPP-B2b mode should be preprocessed to remove outliers and discontinuities. First, the clock offset data are converted to frequency data via epochwise differencing. The outliers are deleted from the time series of frequency data via the quality strategy. The gaps caused by the deletion have been filled with interpolated values to maintain the continuity of the frequency data.

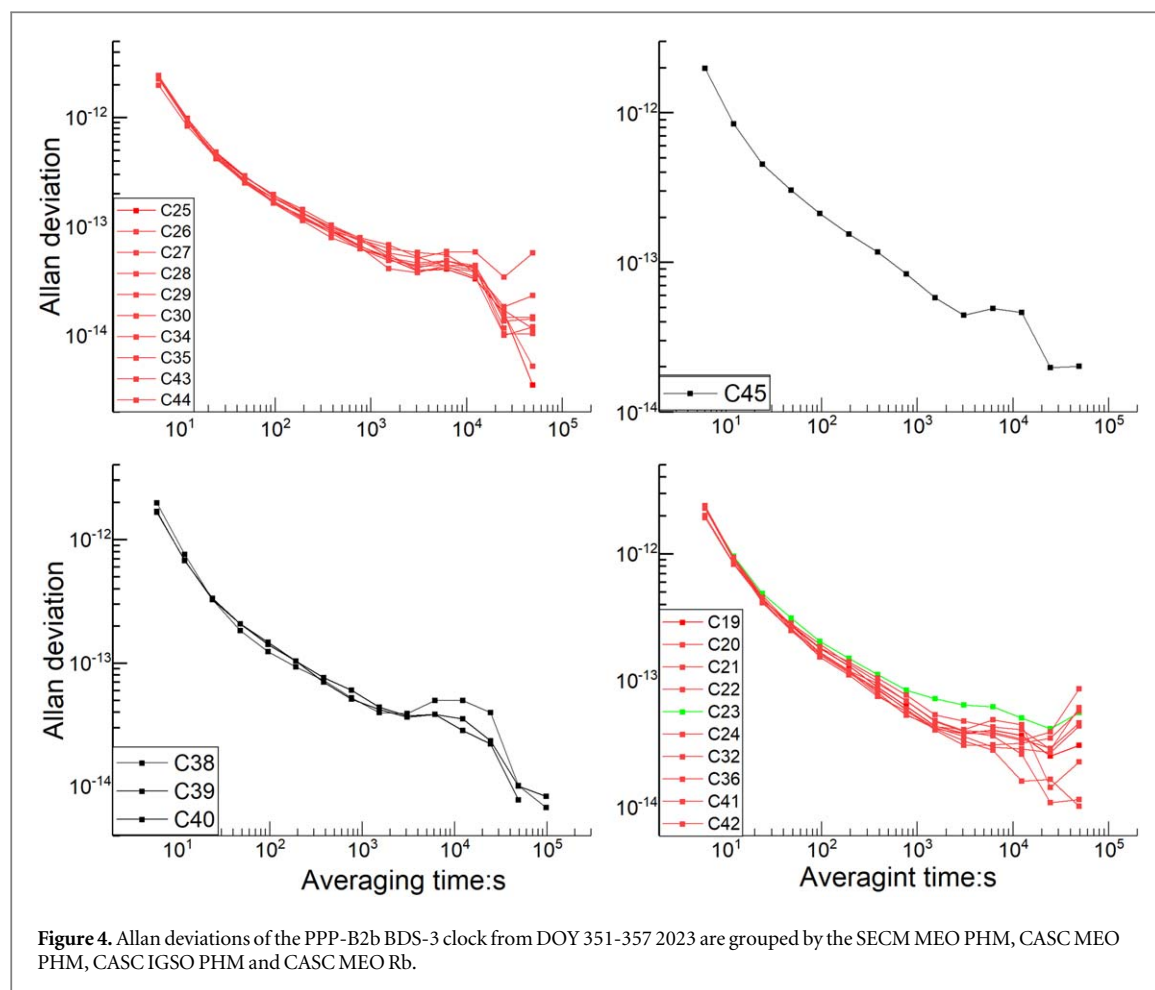


Table 3. BDS-3 and GPS satellite information.

System	Satellite type	Clock type	PRN
BDS-3	SECM, MEO	PHM	C25, C26, C27, C28, C29, C30, C34, C35, C43, C44
	CASC, IGSO	PHM	C38, C39, C40
	CASC, MEO	PHM	C45, C46
	CASC, MEO	Rb	C19, C20, C21, C22, C23, C24, C32, C33, C36, C37, C41, C42
GPS	Block IIR	Rb	G13, G20, G22, G16, G21, G19, G02
	Block IIR-M	Rb	G17, G31, G12, G15, G29, G07, G05
	Block IIF	Rb	G25, G27, G30, G06, G09, G03, G26, G10, G32
	Block IIF	Cs	G08, G24
	Block III	Rb	G4, G18, G23, G14, G11, G28

Figure 4 presents the Allan deviation of all the BDS-3 clocks over averaging time from 6 s to 50000 s. The Allan deviations are grouped according to the satellite clock type and manufacturer in different subplots. We have several findings:

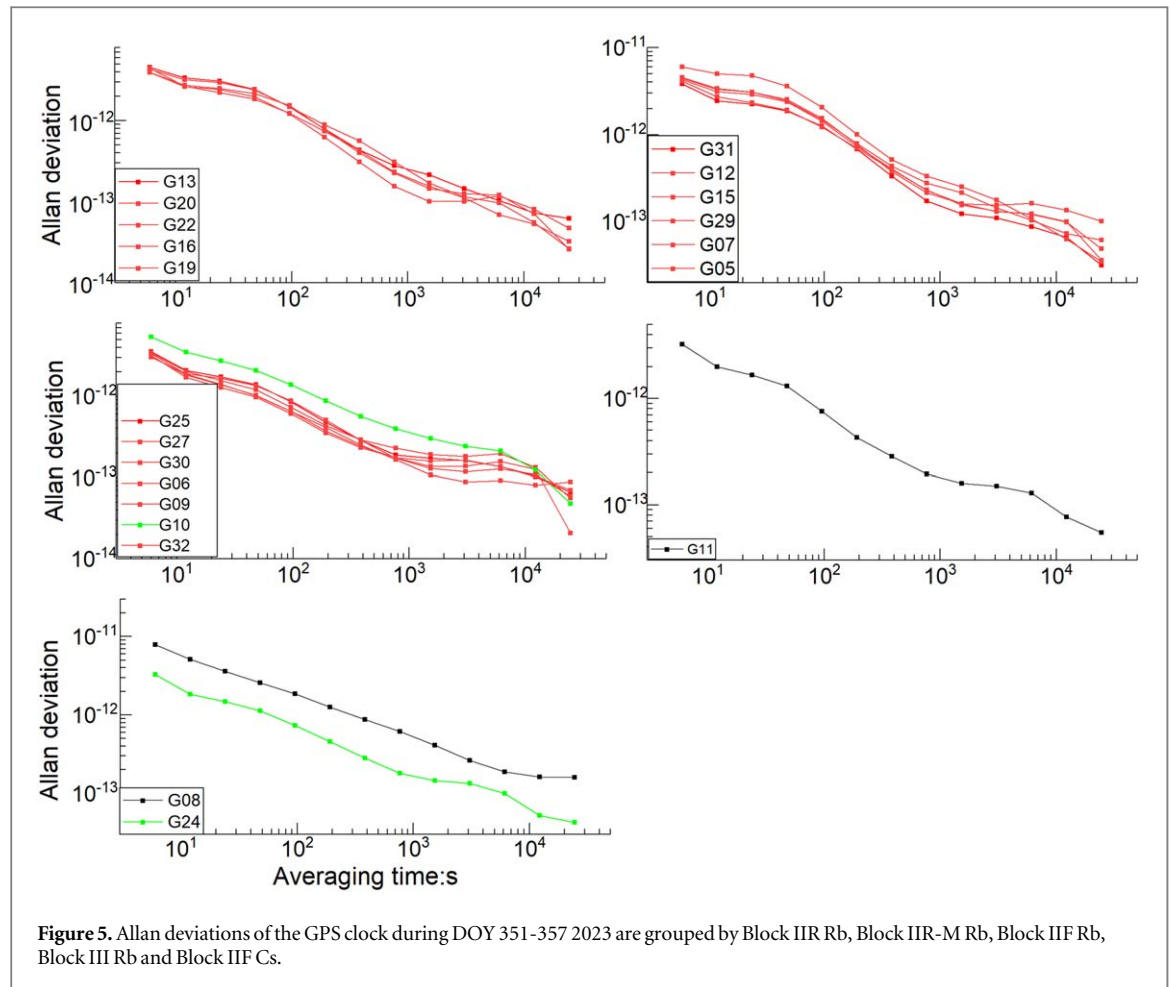
- (1) The average Allan deviations of the SECM MEO PHM, CASC MEO PHM, CASC IGSO PHM and CASC MEO Rb are listed here with different averaging times:

100 s: $1.78\text{e-}13$, $2.12\text{e-}13$, $1.37\text{e-}13$ and $1.74\text{e-}13$;

1000 s: $7.08\text{e-}14$, $8.34\text{e-}14$, $5.46\text{e-}14$ and $6.47\text{e-}14$;

10000 s: $4.14\text{e-}14$, $4.62\text{e-}14$, $3.46\text{e-}14$ and $3.78\text{e-}14$.

For different types of BDS-3 satellites, no obvious discrepancy exists in terms of stability. The stability at 100 s, 1000 s, and 10000 s was approximately $1\sim 2\text{e-}13$, $5\sim 9\text{e-}14$, and $3\sim 5\text{e-}14$, respectively. In terms of the PHM, the stability of the CASC MEO is the worst, whereas that of the CASC IGSO is the best. In terms of CASC, the stability of Rb is slightly better than that of PHM.



- (2) With respect to the four subplots, before 10000 s, the stabilities of the specific satellite clocks agree well. After 10000 s, the stability slightly differs.
- (3) For the CASC IGSO PHM Allan deviation, a bump appears in three IGSO satellites when the averaging time increases. The stability of the IGSO clocks has sharply deteriorated. Concentrating on the performance of the CASC MEO Rb, the stability of C23 (colour in green) is less stable than that of the other satellites.

Now, we turn to figure 5, the Allan deviation of all the GPS clocks from 6 s to 50000 s. The subplots are grouped by satellite type. We have several findings:

- (1) Averaged Allan deviations of Block IIR Rb, Block IIR-M Rb, Block IIF Rb, Block III Rb and Block IIF Cs are listed here with different averaging times:
 - 100 s: $1.39\text{e-}12$, $1.50\text{e-}12$, $8.27\text{e-}13$, $7.57\text{e-}13$ and $1.29\text{e-}12$;
 - 1000 s: $2.40\text{e-}13$, $2.44\text{e-}13$, $2.14\text{e-}14$, $1.95\text{e-}13$ and $4.00\text{e-}14$;
 - 10000 s: $6.66\text{e-}14$, $8.98\text{e-}14$, $1.11\text{e-}13$, $7.69\text{e-}13$ and $1.07\text{e-}13$.
 For different types of GPS satellites, obvious discrepancies exist in terms of stability. The stability of Block IIR Rb and Block IIR-M Rb gradually improve with increasing averaging time, while the 1000 s stability of Block IIF Rb, Block III Rb and Block IIF Cs is worse than that of 100 s and 10000 s. With respect to the five subplots, the stability of G10 and G08 deviates from the overall tendency. The reason is that the fluctuations in the clock offsets worsen the stability. Other satellites of the same clock type agree well.
- (2) Comparably, Block IIF Rb clocks are relatively more stable than other clocks. The stabilities of the Block IIR Rb and Block IIR-M Rb clocks are nearly the same and slightly worse than those of the other clocks.

The points in figure 6 depict the average Allan deviation of PPP-B2b and the GBM clock at several typical averaging times. The symbol square denotes PPP-B2b products, and the symbol circle denotes GBM products. Black denotes the GPS satellite clock, and red denotes the BDS-3 satellite clock.

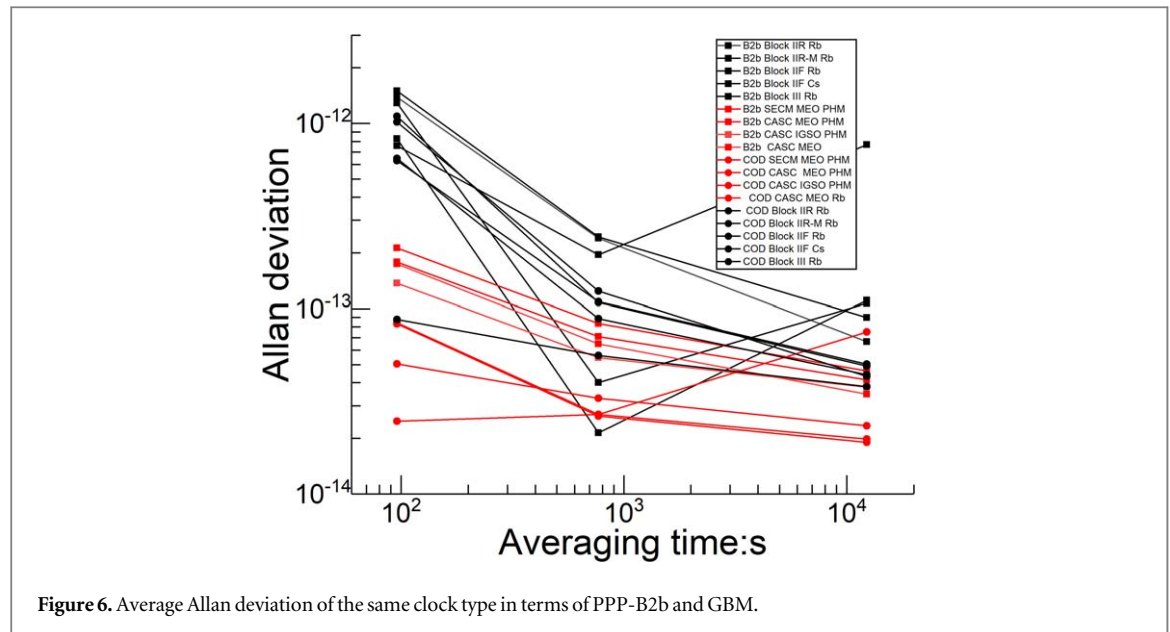


Figure 6. Average Allan deviation of the same clock type in terms of PPP-B2b and GBM.

- (1) For the PPP-B2b products, the stability of the GPS clock is obviously poorer than that of the BDS-3 clock. Similarly, the same conclusion was drawn with the GBM products. In summary, PPP-B2b products are an effective and reasonable way to evaluate the quality of satellite clocks.
- (2) For both the BDS-3 and the GPS products, the performance of the GBM product is superior to that of the B2b product. The GBM product is estimated via the strict postprocessing strategy and the least square method, whereas the PPP-B2b product is estimated via the real-time strategy and Kalman filter strategy.

4.1.2. Satellite clock prediction performance assessment

The excellent satellite clock short-term performance provides the opportunity to decrease the transmission interval. A linear polynomial is employed as the prediction model for analysing the short-term performance of the PPP-B2b clock offsets. The fitting length is fixed to 120 s, and the prediction time is flexible from 6 s to 120 s. By sliding the fitting window forwards, the clock error between the prediction value and the real value is satisfied with different prediction lengths.

The RMSs of BDS-3 and GPS are shown in figures 7–8, and some findings are as follows:

- (1) The tendencies of the prediction accuracies of all the satellites show linear variations. It is reasonable that the prediction accuracy increases as the prediction time increases.
 - (2) The prediction accuracy of C36 is slightly greater than those of the other CASC MEO (Rb) clocks, whereas the prediction accuracies of C34 and C35 are slightly greater than those of the other SECM MEO (PHM) clocks.
 - (3) No obvious deviation exists in the GPS satellite clock prediction accuracy. The different types of GPS satellites agree well.
- (1) In table 4, when the prediction times are 30 s, 60 s and 120 s, the RMSs of the prediction errors of BDS-3 are smaller than 0.01 ns, 0.02 ns, and 0.03 ns, respectively, whereas the RMSs of the prediction errors of the GPS are smaller than 0.28 ns, 0.38 ns, and 0.58 ns, respectively. The prediction error of BDS-3 is better than that of the GPS at all three typical prediction times. Block IIF Cs demonstrates the same prediction accuracy as that of BDS-3. Compared with that of the BDS-3 PHM, which is restricted by the accuracy of the GPS clock offset prediction, the MEO Rb presents a premium prediction accuracy. The suggestion is that PPP-B2b GPS corrections need short-latency estimation and frequent updates. For that, it meets the user requirements for the precise service.
 - (2) The CASC MEO Rb demonstrates considerable prediction accuracy with the SECM MEO PHM. The prediction accuracy of the CASC MEO PHM is significantly greater than that of the SECM MEO PHM.

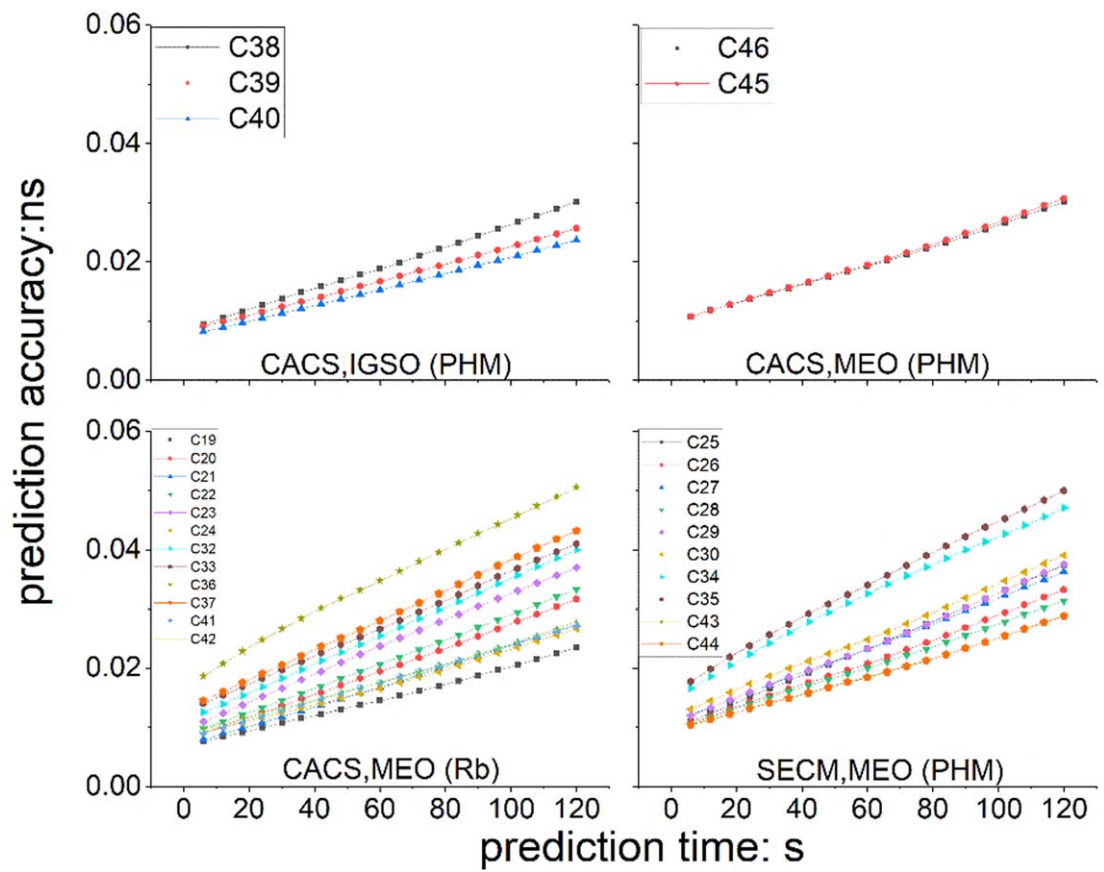


Figure 7. Prediction accuracy of the BDS-3 clock offset.

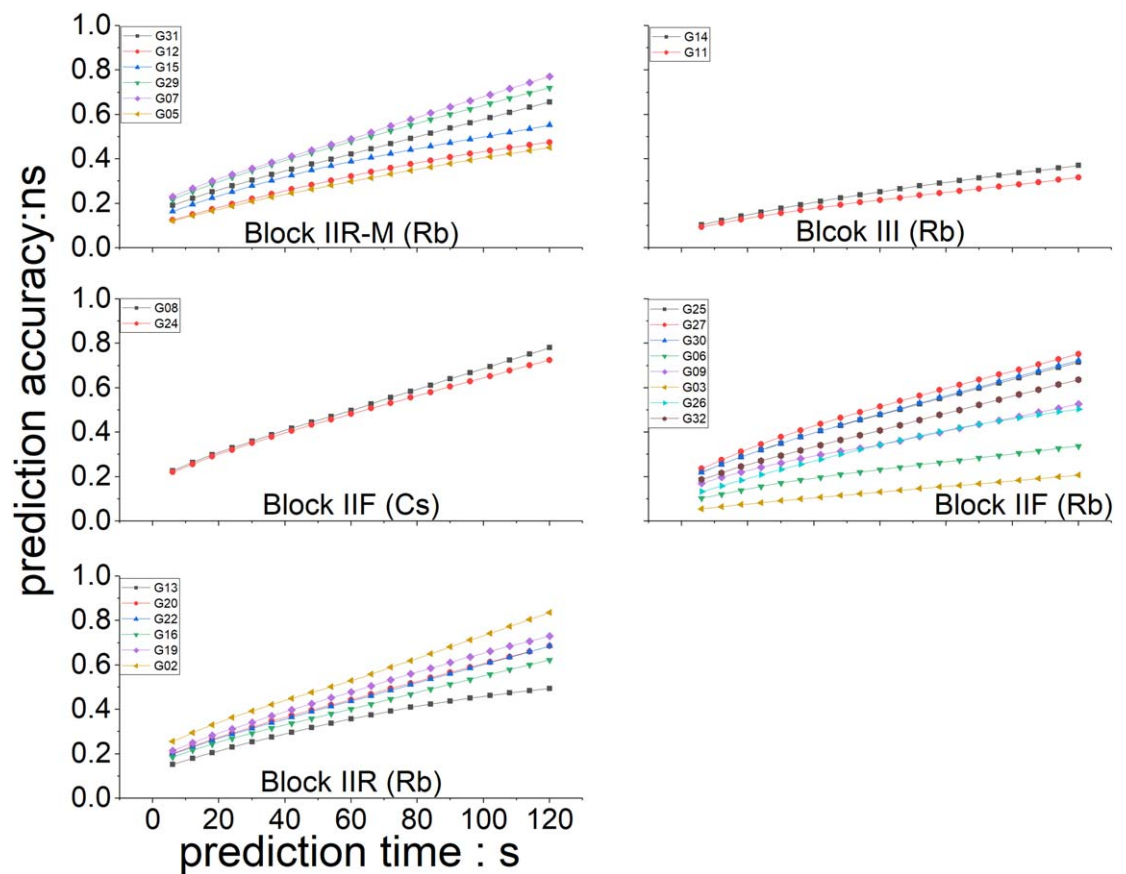


Figure 8. Prediction accuracy of the GPS clock offset.

Table 4. Prediction errors for all kinds of satellite clock offsets.

Satellite type	Prediction error (ns)					
	Prediction time: 30 s		Prediction time: 60 s		Prediction time: 120 s	
	mean	max	mean	max	mean	max
Block III (Rb)	0.167	0.178	0.234	0.252	0.343	0.371
Block IIF (Cs)	0.355	0.360	0.490	0.499	0.753	0.780
Block IIF (Rb)	0.266	0.378	0.367	0.515	0.550	0.752
Block IIR (Rb)	0.319	0.392	0.441	0.529	0.676	0.835
Block IIR-M (Rb)	0.286	0.358	0.400	0.491	0.605	0.771
CACS,IGSO (PHM)	0.013	0.014	0.017	0.019	0.026	0.030
CACS,MEO (PHM)	0.015	0.015	0.019	0.019	0.030	0.031
CACS,MEO (Rb)	0.016	0.027	0.022	0.035	0.034	0.051
SECM,MEO (PHM)	0.018	0.026	0.024	0.034	0.037	0.050

- (3) The prediction accuracy of Block IIR Rb is in accordance with that of Block IIR-M Rb and Block IIF Rb. The prediction accuracies of Block III Rb and Block IIF Cs are significantly better than that of Block II Rb. Focusing on the left panel of figure 8, from Block IIR to Block III, the prediction accuracy of the Rb clock gradually increases as the technique is updated.
- (4) The prediction accuracies of the GPS products are worse than those of the BDS-3 products, and the same conclusion is drawn with respect to the stability of the PPP-B2b products.
- (5) Considering the prediction accuracy of the BDS-3 B2b clock, it is possible to decrease the transmission interval of BDS-3 B2b clock correction.

4.1.3. Satellite clock precision evaluation

The difference between the reference satellite clock offset and other satellite clock offsets is obtained. The influence on the referenced satellite clock should be removed before the precision of the satellite clock can be evaluated. Later, the specific satellite difference between two clock files was obtained, and the RMS was calculated.

In figure 9, the average STD values of the GPS and BDS-3 satellite clocks are approximately 0.20 ns and 0.15 ns, respectively. The precision of the PPP-B2b BDS-3 satellite clock is slightly lower than that of the GPS clock.

5. PPP-B2b time transfer

DF IF combination is usually employed for PPP. The DF IF combination is usually written as:

$$\begin{cases} P_{r,IF_{ij}}^s = \mu_r^s \cdot \Delta X + cdt_{r,IF_{ij}} + m_{r,w}^s \cdot Z_w + \psi_{r,IF_{ij}}^s \\ L_{r,IF_{ij}}^s = \mu_r^s \cdot \Delta X + cdt_{r,IF_{ij}} + m_{r,w}^s \cdot Z_w + N_{r,IF_{ij}}^s + \xi_{r,IF_{ij}}^s \end{cases} \quad (7)$$

$$\begin{cases} cdt_{r,IF_{ij}} = cdt_r + dt_{r,IF_{ij}} \\ dt_{r,IF_{ij}} = \alpha_{ij}d_{r,i} + \beta_{ij}d_{r,j} \\ \alpha_{ij} = \frac{f_i^2}{f_i^2 - f_j^2}; \beta_{ij} = -\frac{f_j^2}{f_i^2 - f_j^2}; i \neq j \end{cases} \quad (8)$$

where $P_{r,IF_{ij}}^s$ and $L_{r,IF_{ij}}^s$ are the OC terms of the IF pseudorange observation and carrier phase observation, respectively; s and r denote the satellite and receiver, respectively; Δx denotes the coordinate increment; μ_r^s is the coefficient vector for coordinates; $d_{r,i}$ and $d_{r,j}$ denote the hardware delay of the receiver at frequencies i and j, respectively; dt_r is the receiver clock offset; $dt_{r,IF_{ij}}$ is the receiver clock offset of the IF combination, including the hardware delay; $d_{r,IF_{ij}}$ is the hardware delay of the IF combination; Z_w is the tropospheric ZWD (Zenith Wet Delay); $m_{r,w}^s$ is the wet mapping function; $N_{r,IF_{ij}}^s$ is the float IF ambiguity; and $\psi_{r,IF_{ij}}^s$ and $\xi_{r,IF_{ij}}^s$ are the IF pseudorange noise and carrier phase noise, respectively. The receiver clock offset is not influenced by hardware delay because of constant characteristics.

In the PPP-B2b service, the time datum of the BDS-3 satellite clock is aligned with the BDT, whereas the GPS satellite clock is not. The influence on the time datum can be removed because of the differential mode of time

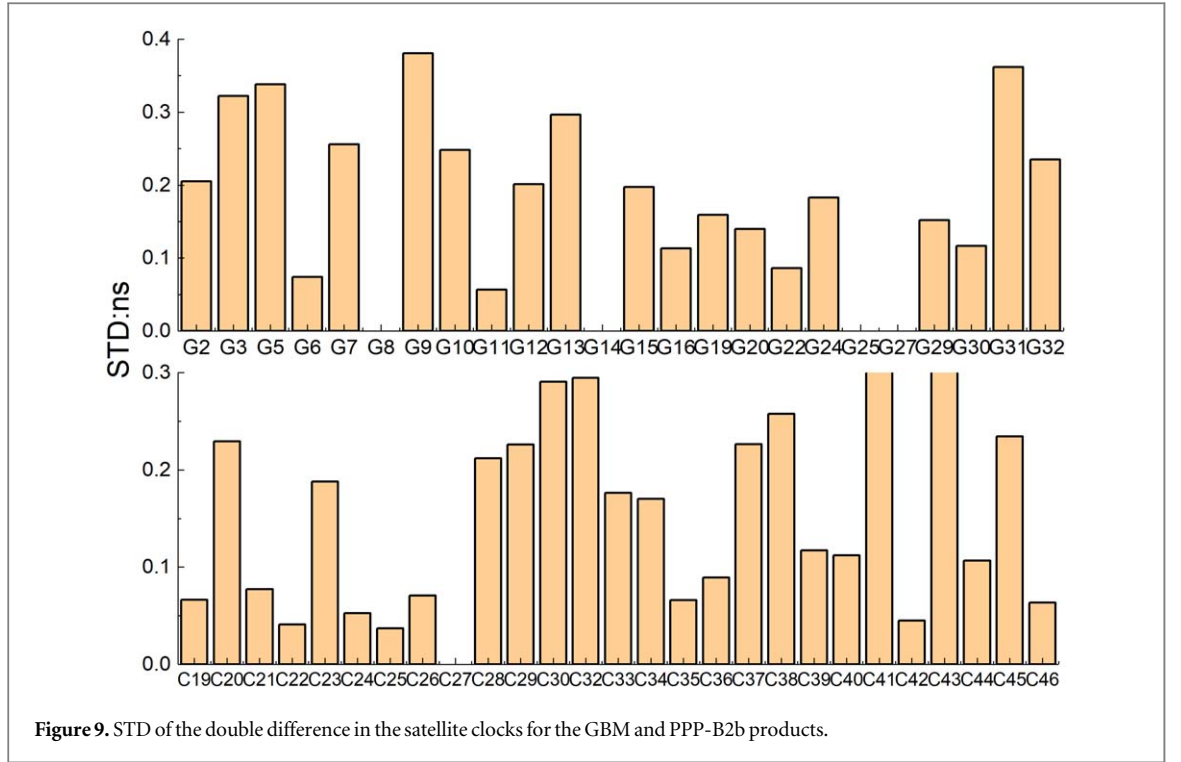


Figure 9. STD of the double difference in the satellite clocks for the GBM and PPP-B2b products.

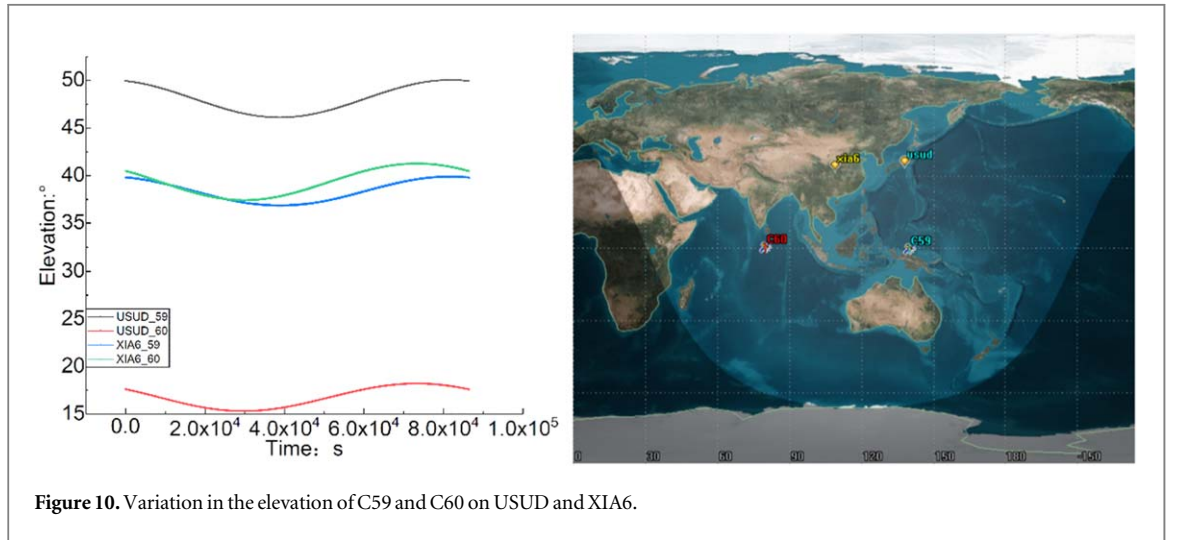


Figure 10. Variation in the elevation of C59 and C60 on USUD and XIA6.

transfer. The difference $dt_{r,IF_{ij}}$ between the local time and BDT can be described as

$$dt_{r,IF_{ij}} = T_{local} - BDT \quad (9)$$

where T_{local} represents the local time of the users. The difference in the time transfer ΔT is expressed as:

$$\Delta T = dt_{r1,IF_{ij}} - dt_{r2,IF_{ij}} = T_{local,1} - BDT - (T_{local,2} - BDT) = T_{local,1} - T_{local,2} \quad (10)$$

where $T_{local,1}$ and $T_{local,2}$ are the local times of different users.

Stations capable of connecting the atomic clock and receiving PPP-B2b signals are scarce in the Asia-Pacific region. Two stations are selected to participate in the experiment, namely, USUD and XIA6. USUD is equipped with a Caesium clock, whereas XIA6 is connected to UTC (NTSC).

The observations cover 6 days from DOY 351 to 356, 2023.

Figure 10 shows the variations in the elevations of C59 and C60 at USUD and XIA6. For station XIA6, the elevation evolution of C59 and C60 is nearly the same. It can be explained that XIA6 was located in the middle of C59 and C60. For station USUD, the elevation of C60 is obviously lower than that of C59. Therefore, the transmission mode of multiple satellites can ensure that the B2b service is maximal. The processing strategy is listed in table 5.

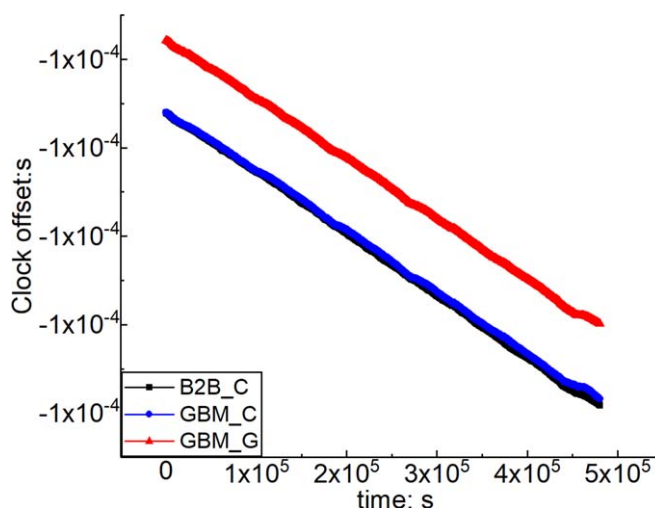


Figure 11. XIA6-USUD clock offset obtained from GPS/BDS PPP via GBM and B2b products.

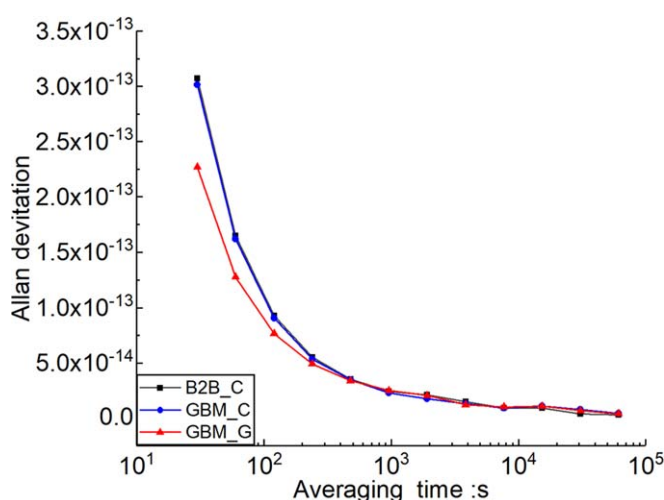


Figure 12. XIA6-USUD ADEV of time transfer using GBM and B2b products.

Table 5. BDS-3 PPP processing strategy.

Estimator	Kalman filter
Signals	BDS-3: B1I/B3I GPS: L1/L2
PCV and PCO	igs14.atx
Precise products	PPP-B2b correction GBM final product
Phase ambiguities	Estimated as a constant at each arc
Tro	Estimated as a random walk process
Receiver clock offset	Estimated with a white noise model

Figure 11 presents XIA6-USUD clock offsets obtained from GPS/BDS PPP via different products. The clock offset of BDS PPP-B2b agrees well with the clock offset of BDS PPP-GBM: first, it indicates the availability of B2b products; second, it presents the potential for BDS PPP-B2b time transfer. The tendency of the BDS clock offset is nearly the same as that of the GPS clock offset, whereas a stable bias appears in the clock offsets of the GPS and BDS clocks. This finding is attributed to the discrepancy of different navigation systems. To explore the performance of BDS PPP-B2b further, with GPS PPP as a reference, the standard deviations (STDs) of BDS PPP-B2b and BDS PPP-GBM are 0.50 ns and 0.45 ns, respectively. The STD of BDS PPP-B2b is slightly poorer than that of BDS PPP-GBM. This benefits the strict processing strategy of the GBM final products. Figure 12 presents

the XIA6-USUD time link ADEV for three types of products. The stability of BDS PPP-B2b time transfer is almost the same as that of BDS PPP-GBM at each averaging time. For shorter averaging times, the stability of BDS PPP-B2b is less stable than that of GPS PPP before 960 s and is equivalent to that of GPS PPP at longer averaging times. The frequency stabilities of several important averaging times are listed here: the frequency stabilities of BDS PPP-B2b, BDS PPP-GBM and GPS PPP-GBM are approximately $9.25\text{E-}14$, $9.05\text{E-}14$, and $7.64\text{E-}14$, respectively, at 100 s; approximately $2.42\text{E-}14$, $2.30\text{E-}14$, and $2.49\text{E-}14$, respectively, at 1000 s; and approximately $9.34\text{E-}15$, $9.33\text{E-}15$, and $1.02\text{E-}14$, respectively, at 10000 s. The conclusion of frequency stability is similar to the precision of time transfer.

6. Summary

The transmission of PPP-B2b correction provides an internet-independent way to achieve BDS-3 PPP in the Asia-Pacific region. The coverage of PPP-B2b is analysed in this study, which provides guidelines for PPP-B2b users. PPP-B2b corrections were collected for up to seven days, after which the samples were recovered to the precise product. Then, the PPP-B2b clock performance is evaluated, and the possibility of decreasing the transmission interval is discussed. Later, an experiment was conducted to validate the time transfer performance.

In terms of the orbit, the mean STD of the B2b GPS orbit errors is approximately (0.14, 0.74, and 0.57) m along the RAC direction. The mean STD of the B2b BDS-3 orbit errors is approximately 0.09, 0.62, and 0.57 m along the RAC direction. With respect to the satellite clock, the stability of the B2b GPS clock is obviously poorer than that of the BDS-3 clock. In terms of the BDS-3 PHM, the stability of the CASC MEO is the worst, whereas that of the CASC IGSO is the best. The mean stabilities of the CASC IGSO PHM at 100 s, 1000 s and 10000 s are $1.37\text{E-}13$, $5.46\text{E-}14$ and $3.46\text{E-}14$, respectively. The mean stabilities of the CASC MEO PHM at 100 s, 1000 s and 10000 s are $2.12\text{E-}13$, $8.34\text{E-}14$ and $4.62\text{E-}14$, respectively. Before 10000 s, the stabilities of the specific satellite clocks agree well. After 10000 s, the stability slightly differed. Block IIF Rb clocks are relatively more stable than other clocks. The stabilities of Block IIR Rb and Block IIR-M Rb are nearly the same and slightly worse than those of the other clocks. For satellite clock prediction performance, when the prediction time reaches 120 s, the RMS prediction error of BDS-3 is 0.03 ns. The RMS of the prediction error of the GPS is smaller than 0.28 ns when the prediction time is up to 30 s. This method is capable of decreasing the transmission interval of the BDS-3 B2b clock correction to transmit other information. The precision of time transfer obtained from BDS-B2b PPP (0.50 ns) is less than that of BDS-GBM PPP (0.45 ns). The frequency stability of BDS-3 PPP is approximately $9.25\text{E-}14$, $2.42\text{E-}14$, and $9.34\text{E-}15$ at 100 s, 1000 s, and 10000 s, respectively.

Data availability statement

All data that support the findings of this study are included within the article (and any supplementary files).

Author contributions

NConceptualization, methodology and validation: Qin Weijin; PPP-B2b recovery: Zhang Zhe; orbit analysis: Su Hang; and discussion—Wei Pei and Yang Xuhai. All the authors have read and agreed to the published version of the manuscript.

Funding

This work was performed with the support of the Advanced Strategic Science and Technology Pilot Project (Grant No. XDA0350305), the National Natural Science Foundation of China (Grant No. 12273047), the CAS 'Light of West China' Program (Grant No. xbzg-zdsys-202308) and the Youth Promotion Committee of the Chinese Academy of Science (Grant No. Y2023108).

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